

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VII

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Class : SYBSCIT	Batch : 1
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Hash Table:

Write a program to:

- Implement a hash table with separate chaining for collision handling.

Code:

```
File Edit Format Run Options Window Help
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)].append((key,value))

insert(10,"Nidhi")
insert(15,"Mehek")
insert(20,"Vedant")
insert(32,"Preesha")
insert(33,"Disha")

print(table)
```

Output:

```
==== RESTART: C:/Users/nidhi/Downloads/Nidhi DS/prac7.py ====
[[ (10, 'Nidhi'), (15, 'Mehek'), (20, 'Vedant')], [], [(32, 'Preesha')], [(33, 'Disha')], []]
```

b. Store and retrieve data from the hash table.

Code:

```
File Edit Format Run Options Window Help
size = 5
table = [[] for _ in range(size)]

def insert (key,value):
    table[(key%size)].append((key,value))

def search(key):
    for k,v in table[key%size]:
        if k==key:
            return v
    return "Not found"

insert(10,"Nidhi")
insert(15,"Mehek")
insert(20,"Vedant")
insert(32,"Preesha")
insert(33,"Disha")

print(table)

key = int(input("\nEnter Key: "))

print(search(key))
```

Output:

```
===== RESTART: C:/Users/nidhi/Downloads/Nidhi DS/prac7.py =====
[[ (10, 'Nidhi'), (15, 'Mehek'), (20, 'Vedant') ], [], [(32, 'Preesha')], [(33, 'Disha')], []]

Enter Key: 10
Nidhi
```

